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EDS

CodeTantra Codes

Name: Aniket A.D. Avdhut

PRN: 202501040048

Division: CS1

Batch: C13

Subject: Essentials of Data Science

Practical 4

4.1. Practice Lab Assignments

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4.1.3. Student Information

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4.2.5. Titanic Dataset Analysis and Data Cleaning

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4.2.8. Titanic Dataset Analysis and Data Cleaning - 4

4.1.1. Pandas - series creation and manipulation

Write a Python program that takes a list of numbers from the user, creates a Pandas series from it, and then calculates the mean of even and odd numbers separately using the `groupby` and `mean()` operations.

Hint: You can group the Series based on whether each number is even or odd.

Input Format:

- The user should enter a list of numbers separated by space when prompted.

Output Format:

- The program should display the mean of even and odd numbers separately.
- Each mean value should be displayed with a label indicating whether it corresponds to even or odd numbers.

Refer to the sample test cases for better understanding regarding input and output format.

Sample Test Cases

seriesMa...

```
1 import pandas as pd
2
3 # Take inputs from the user to create a list of numbers
4 numbers = list(map(int, input().split()))
5
6 # Create a Pandas series from the list of numbers
7 series = pd.Series(numbers)
8 # Grouping by even and odd numbers and calculating the mean
9 grouped = series.groupby(series%2==0).mean()
10
11 # Display the mean of even and odd numbers with labels
12 grouped.index = ['Even' if is_even else 'Odd' for is_even in grouped.index]
13 print("Mean of even and odd numbers:")
14 print(grouped)
```

Terminal Test cases

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4.1.2. Student Information

Write a program to read a text file containing student information (name, age, and grade) using Pandas. Perform the following tasks:

- Display the first five rows of the data frame.
- Calculate the average age of the students (limit the average age up to 2 decimal places).
- Filter out the students who have a grade above a certain threshold (consider the threshold grade is 'B').

Note:

Refer to the displayed test cases for better understanding.

Sample Test Cases

studentin...

```
1 import pandas as pd
2
3 # Read the text file into a DataFrame
4 file = input()
5 data = pd.read_csv(file, sep=";", header=None, names=["Name", "Age", "Grade"])
6
7
8 # write your code here..
9 print('First five rows:')
10 print(data.head())
11
12 avg_age = data['Age'].mean()
13 print(f"Average age: {round(avg_age, 2)}")
14
15 filter_students = data[data['Grade'] <= 'B']
16 print("Students with a grade up to B")
17 print(filter_students)
```

Terminal Test cases

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4.1.2. Dictionary to dataframe

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A dictionary of lists has been provided to you in the editor. Create a DataFrame from the dictionary of lists and perform the listed operations, then display the DataFrame before and after each manipulation.

Create the DataFrame:

- Convert the dictionary to a Pandas DataFrame.

Add a new row:

- Take inputs from the user for the new row data (name, age).
- Add the new row to the DataFrame.
- Display the DataFrame after adding the new row.

Modify a row:

- Modify a specific row by changing the age. Take the row index and new age value from the user.
- Display the DataFrame after modifying the row.

Delete a row:

- Take the row index to be deleted from the user.
- Remove the specified row.
- Display the DataFrame after deleting the row.

Add a new column:

- Add a column **Gender** with values taken from the user.
- Display the DataFrame after adding the new column.

Modify a column:

- Convert names to uppercase.
- Display the DataFrame after modifying the column.

Delete a column:

- Remove the **Age** column.
- Display the DataFrame after deleting the column.

Sample Test Cases

A dictionary of lists has been provided to you in the editor. Create a DataFrame from the dictionary of lists and perform the listed operations, then display the DataFrame before and after each manipulation.

Create the DataFrame:

- Convert the dictionary to a Pandas DataFrame.

Add a new row:

- Take inputs from the user for the new row data (name, age).
- Add the new row to the DataFrame.
- Display the DataFrame after adding the new row.

Modify a row:

- Modify a specific row by changing the age. Take the row index and new age value from the user.
- Display the DataFrame after modifying the row.

Delete a row:

- Take the row index to be deleted from the user.
- Remove the specified row.
- Display the DataFrame after deleting the row.

Add a new column:

- Add a column **Gender** with values taken from the user.
- Display the DataFrame after adding the new column.

Modify a column:

- Convert names to uppercase.
- Display the DataFrame after modifying the column.

Delete a column:

- Remove the **Age** column.
- Display the DataFrame after deleting the column.

Sample Test Cases

datafram...

Submit

```
1 import pandas as pd
2
3 # Provided dictionary of lists
4 data = {
5     'Name': ['Alice', 'Bob', 'Charlie'],
6     'Age': [25, 30, 35],
7 }
8
9 # Convert the dictionary to a DataFrame
10 df = pd.DataFrame(data)
11
12 # Display the original DataFrame
13 print("Original DataFrame:")
14 print(df)
15
16 # Adding a new row
17 new_name=input("New name: ")
18 new_age=int(input("New age: "))
19 df.loc[len(df)]=[new_name,new_age]
20
21 # Display the DataFrame after adding a new row
22 print("After adding a row:\n",df)
23
24 # Modifying a row
25 row_modify = int(input("Index of row to modify: "))
26 modify_age = int(input("New age: "))
27 df.at[row_modify,'Age'] = modify_age
28
29 # Display the DataFrame after modifying a row
30 print("After modifying a row:")
31 print(df)
32
33 # Deleting a row
34 row_delete=int(input("Index of row to delete: "))
35 df= df.drop(index=row_delete).reset_index(drop=True)
```

Terminal Test cases

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Exec

```
32
33 # Deleting a row
34 row_delete=int(input("Index of row to delete: "))
35 df= df.drop(index=row_delete).reset_index(drop=True)
36
37 # Display the DataFrame after deleting a row
38 print("After deleting a row:")
39 print(df)
40
41 # Adding a new column
42 gender=input("Enter genders separated by space: ").split()
43 df["Gender"]=gender
44
45 # Display the DataFrame after adding a new column
46 print("After adding a new column:")
47 print(df)
48
49 # Modifying a column
50 df['Name']=df['Name'].str.upper()
51
52 # Display the DataFrame after modifying a column
53 print("After modifying a column:")
54 print(df)
55
56 # Deleting a column
57 df=df.drop(columns='Age').reset_index(drop=True)
58
59 # Display the DataFrame after deleting a column
60 print("After deleting a column:")
61 print(df)
62
63
64
65
66
```

Terminal Test cases

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4.2.1. Month with the Highest Total Sales

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by Month and calculate the total sales for each month.
- Find the month with the highest total sales and display it.
- Also, display the total sales for the best month.

Sample Data:

```
Date,Product,Quantity,Price,City
2025-01-01,Product A,5,20,New York
2025-01-01,Product B,3,15,Los Angeles
2025-01-02,Product A,7,20,New York
2025-01-02,Product C,4,30,Chicago
2025-01-03,Product B,2,15,Chicago
2025-01-03,Product A,8,20,Los Angeles
2025-01-04,Product C,6,30,New York
2025-01-04,Product B,5,15,Los Angeles
2025-01-05,Product A,3,20,Chicago
2025-01-05,Product C,10,30,Los Angeles
```

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Sample Test Cases

monthFor...

```
1 import pandas as pd
2
3 # Prompt the user for the file name
4 file_name = input()
5
6 # Load the data
7 df = pd.read_csv(file_name)
8
9 df["Total Sales"] = df["Quantity"] * df["Price"]
10 df["Date"] = pd.to_datetime(df["Date"])
11 df["Month"] = df["Date"].dt.to_period('M')
12
13 monthly_sales = df.groupby("Month")["Total Sales"].sum()
14
15 # Find the month with the highest total sales
16 best_month = monthly_sales.idxmax()
17 highest_sales = monthly_sales.max()
18
19 print(f"Best month: {best_month}")
20 print(f"Total sales: ${highest_sales:.2f}")
21
```

4.2.2. Best Selling Product

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Find the product that sold the most in terms of quantity sold.
- Display the product that sold the most and the total quantity sold for that product.

Sample Data:

```
Date,Product,Quantity,Price,City
2025-01-01,Product A,5,20,New York
2025-01-01,Product B,3,15,Los Angeles
2025-01-02,Product A,7,20,New York
2025-01-02,Product C,4,30,Chicago
2025-01-03,Product B,2,15,Chicago
2025-01-03,Product A,8,20,Los Angeles
2025-01-04,Product C,6,30,New York
2025-01-04,Product B,5,15,Los Angeles
2025-01-05,Product A,3,20,Chicago
2025-01-05,Product C,10,30,Los Angeles
```

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Sample Test Cases

monthFor...

```
1 import pandas as pd
2
3 # Prompt the user for the file name
4 file_name = input()
5
6 # Load the data
7 df = pd.read_csv(file_name)
8
9 product_quantity = df.groupby('Product')['Quantity'].sum()
10 # Find the product with the highest total quantity sold
11 best_product = product_quantity.idxmax()
12 highest_quantity = product_quantity.max()
13
14 # Display the result
15 print(f"Best selling product: {best_product}")
16 print(f"Total quantity sold: {highest_quantity}")
17
```

4.2.3. City that Sold the Most Products

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by city and calculate the total quantity of products sold for each city.
- Find the city that sold the most products (based on the total quantity sold).

Sample Data:

```
Date,Product,Quantity,Price,City
2025-01-01,Product A,5,20,New York
2025-01-01,Product B,3,15,Los Angeles
2025-01-02,Product A,7,20,New York
2025-01-02,Product C,4,30,Chicago
2025-01-03,Product B,2,15,Chicago
2025-01-03,Product A,8,20,Los Angeles
2025-01-04,Product C,6,30,New York
2025-01-04,Product B,5,15,Los Angeles
2025-01-05,Product A,3,20,Chicago
2025-01-05,Product C,10,30,Los Angeles
```

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Sample Test Cases

monthFor...

```
1 import pandas as pd
2
3 # Prompt the user for the file name
4 file_name = input()
5
6 # Load the data
7 df = pd.read_csv(file_name)
8
9 city_sales = df.groupby("City")["Quantity"].sum()
10
11 best_city = city_sales.idxmax()
12
13 # Display the result
14 print(f"City sold the most products: {best_city}")
15
```

4.2.4. Most Frequently Sold Product Pairs

21:30

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the following columns: Date, Product, Quantity, Price, and City
- For each date, find all pairs of products that were sold together (i.e., two products sold on the same date).
- Output the product pairs that was sold most frequently.

Sample Data:

```
Date,Product,Quantity,Price,City
2025-01-01,Product A,5,20,New York
2025-01-01,Product B,3,15,Los Angeles
2025-01-02,Product A,7,20,New York
2025-01-02,Product C,4,30,Chicago
2025-01-03,Product B,2,15,Chicago
2025-01-03,Product A,8,20,Los Angeles
2025-01-04,Product C,6,30,New York
2025-01-04,Product B,5,15,Los Angeles
2025-01-05,Product A,3,20,Chicago
2025-01-05,Product C,10,30,Los Angeles
```

Explanation:

Transactions:

- 2025-01-01: Product A, Product B
- 2025-01-02: Product A, Product C
- 2025-01-03: Product B, Product A
- 2025-01-04: Product C, Product B
- 2025-01-05: Product A, Product C

Now, let's count how often the pairs of products appear together:

- Product A and Product B: Appear in transactions on 2025-01-01 and 2025-01-03.
- Product A and Product C: Appear in transactions on 2025-01-02 and 2025-01-05.
- Product B and Product C: Appear in transactions on 2025-01-04.

Most Frequent Product Combinations:

- Product A and Product B (2 times)
- Product A and Product C (2 times)

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

frequent...

```
1 import pandas as pd
2 from itertools import combinations
3 from collections import Counter
4
5 # Prompt user to input the file name
6 file_name = input()
7
8 # Read data from the specified CSV file
9 df = pd.read_csv(file_name)
10
11 # 1. Group the data by 'Date' and get the list of products for each date
12 daily_transactions = df.groupby('Date')['Product'].apply(list)
13 # Initialize a Counter to store the frequency of each pair
14 pair_counts = Counter()
15
16 # 2. Iterate through each date's transaction
17 for products in daily_transactions:
18     products = sorted(products)
19     pairs = list(combinations(products, 2))
20     pair_counts.update(pairs)
21
22 # 3. Find the maximum frequency count and print all pairs that match this maximum frequency
23 if pair_counts:
24     max_count = max(pair_counts.values())
25     for pair, count in pair_counts.items():
26         if count == max_count:
27             print(f'{pair[0]} and {pair[1]}: {count} times')
28 else:
29     print("No product pairs found.")
```

Terminal Test cases

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4.2.5. Titanic Dataset Analysis and Data Cleaning

14:23

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset. For each question, perform necessary data cleaning, transformations, and calculations as required.

- Display the first 5 rows of the dataset.
- Display the last 5 rows of the dataset.
- Get the shape of the dataset (number of rows and columns).
- Get a summary of the dataset (using `.info()`).
- Get basic statistics (mean, standard deviation, etc.) of the dataset using `.describe()`.
- Check for missing values and display the count of missing values for each column.
- Fill missing values in the 'Age' column with the median age.
- Fill missing values in the 'Embarked' column with the most frequent value (`mode()`).
- Drop the 'Cabin' column due to many missing values.
- Create a new column, 'FamilySize' by adding the 'SibSp' and 'Parch' columns.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,S
6,0,3,"Moran, Mr. James",male,0,0,330877,8.4583,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,F46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,1,1,349909,21.075,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,C
```

titanicDat...

```
1 import pandas as pd
2 import numpy as np
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # 1. Display the first 5 rows of the dataset
8 print(data.head())
9 # 2. Display the last 5 rows of the dataset
10 print(data.tail())
11 # 3. Get the shape of the dataset
12 print(data.shape)
13 # 4. Get a summary of the dataset (info)
14 print(data.info())
15 # 5. Get basic statistics of the dataset
16 print(data.describe())
17 # 6. Check for missing values
18 print(data.isnull().sum())
19 # 7. Fill missing values in the 'Age' column with the median age
20 data['Age'].fillna(data['Age'].median(), inplace=True)
21 # 8. Fill missing values in the 'Embarked' column with the mode
22 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
23 # 9. Drop the 'Cabin' column due to many missing values
24 data.drop(columns='Cabin', inplace=True)
25 # 10. Create a new column 'FamilySize' by adding 'SibSp' and 'Parch'
26 data['FamilySize'] = data['SibSp'] + data['Parch']
27
```

Terminal Test cases

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4.2.6. Titanic Dataset Analysis and Data Cleaning - 2

13:30

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

- Create a new column 'IsAlone' which is 1 if the passenger is alone (FamilySize = 0), otherwise 0.
- Convert the 'Sex' column to numeric values (male: 0, female: 1).
- One-hot encode the 'Embarked' column, dropping the first category.
- Get the mean age of passengers.
- Get the median fare of passengers.
- Get the number of passengers by class.
- Get the number of passengers by gender.
- Get the number of passengers by survival status.
- Calculate the survival rate of passengers.
- Calculate the survival rate by gender.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,S
6,0,3,"Moran, Mr. James",male,0,0,330877,8.4583,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,F46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,1,1,349909,21.075,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,C
```

Note: Refer to the visible test case for better reference.

titanicDat...

```
1 import pandas as pd
2 import numpy as np
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6 data['FamilySize'] = data['SibSp'] + data['Parch']
7
8 # Create a new column 'IsAlone' which is 1 if the passenger is alone (i.e., FamilySize = 0),
9 otherwise 0
10 data['IsAlone'] = np.where(data['FamilySize'] == 0, 1, 0)
11 # data['IsAlone'] = (data['FamilySize'] == 0).astype(int) # Alternate way
12 # 2. Convert 'Sex' to numeric (male: 0, female: 1)
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 # 3. One-hot encode the 'Embarked' column
15 data = pd.get_dummies(data, columns=['Embarked'])
16 # 4. Get the mean age of passengers
17 mean_age = data['Age'].mean()
18 print(mean_age)
19 # 5. Get the median fare of passengers
20 median_fare = data['Fare'].median()
21 print(median_fare)
22 # 6. Get the number of passengers by class
23 print(data['Pclass'].value_counts())
24 # 7. Get the number of passengers by gender
25 print(data['Sex'].value_counts())
26 # 8. Get the number of passengers by survival status
27 print(data['Survived'].value_counts())
28 # 9. Calculate the overall survival rate
29 survival_rate = data['Survived'].mean()
30 print(format(survival_rate))
31 # 10. Calculate the survival rate by gender
32 survival_by_gender = data.groupby('Sex')['Survived'].mean()
33 print(survival_by_gender)
```

Terminal Test cases

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4.2.7. Titanic Dataset Analysis and Data Cleaning - 3

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Calculate the survival rate by class.
2. Calculate the survival rate by embarkation location (Embarked_S).
3. Calculate the survival rate by family size (FamilySize).
4. Calculate the survival rate by being alone (IsAlone).
5. Get the average fare by passenger class (Pclass).
6. Get the average age by passenger class (Pclass).
7. Get the average age by survival status (Survived).
8. Get the average fare by survival status (Survived).
9. Get the number of survivors by class (Pclass).
10. Get the number of non-survivors by class (Pclass).

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,0,0,330877,8.4583,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C
```

```
titanicDat...
1 import pandas as pd
2 import numpy as np
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6 data['FamilySize'] = data['SibSp'] + data['Parch']
7 data['IsAlone'] = np.where(data['FamilySize'] > 0, 0, 1)
8 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
9
10 # 1. Calculate the survival rate by class
11 print(data.groupby('Pclass')['Survived'].mean())
12 # 2. Calculate the survival rate by embarkation location (Embarked_S)
13 print(data.groupby('Embarked_S')['Survived'].mean())
14 # 3. Calculate the survival rate by family size
15 print(data.groupby('FamilySize')['Survived'].mean())
16 # 4. Calculate the survival rate by being alone
17 print(data.groupby('IsAlone')['Survived'].mean())
18 # 5. Get the average fare by passenger class
19 print(data.groupby('Pclass')['Fare'].mean())
20 # 6. Get the average age by passenger class
21 print(data.groupby('Pclass')['Age'].mean())
22 # 7. Get the average age by survival status
23 print(data.groupby('Survived')['Age'].mean())
24 # 8. Get the average fare by survival status
25 print(data.groupby('Survived')['Fare'].mean())
26 # 9. Get the number of survivors by class
27 # Filter for Survived=1, then count occurrences of Pclass
28 print(data[data['Survived'] == 1]['Pclass'].value_counts())
29 # 10. Get the number of non-survivors by class
30 # Filter for Survived=0, then count occurrences of Pclass
31 print(data[data['Survived'] == 0]['Pclass'].value_counts())
32
33
```

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4.2.8. Titanic Dataset Analysis and Data Cleaning - 4

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Get the number of survivors by gender (Sex).
2. Get the number of non-survivors by gender (Sex).
3. Get the number of survivors by embarkation location (Embarked_S).
4. Get the number of non-survivors by embarkation location (Embarked_S).
5. Calculate the percentage of children (Age < 18) who survived.
6. Calculate the percentage of adults (Age >= 18) who survived.
7. Get the median age of survivors.
8. Get the median age of non-survivors.
9. Get the median fare of survivors.
10. Get the median fare of non-survivors.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,0,0,330877,8.4583,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C
```

```
titanicDat...
1 import pandas as pd
2 import numpy as np
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
7
8
9 # 1. Get the number of survivors by gender
10 print(data[data['Survived'] == 1]['Sex'].value_counts())
11 # 2. Get the number of non-survivors by gender
12 print(data[data['Survived'] == 0]['Sex'].value_counts())
13 # 3. Get the number of survivors by embarkation location
14 print(data[data['Survived'] == 1]['Embarked_S'].value_counts())
15 # 4. Get the number of non-survivors by embarkation location
16 print(data[data['Survived'] == 0]['Embarked_S'].value_counts())
17 # 5. Calculate the percentage of children (Age < 18) who survived
18 print(data[data['Age'] < 18]['Survived'].mean())
19 # 6. Calculate the percentage of adults (Age >= 18) who survived
20 print(data[data['Age'] >= 18]['Survived'].mean())
21 # 7. Get the median age of survivors
22 print(data[data['Survived'] == 1]['Age'].median())
23 # 8. Get the median age of non-survivors
24 print(data[data['Survived'] == 0]['Age'].median())
25 # 9. Get the median fare of survivors
26 print(data[data['Survived'] == 1]['Fare'].median())
27 # 10. Get the median fare of non-survivors
28 print(data[data['Survived'] == 0]['Fare'].median())
29
```

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